Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18RA53]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

PRINCIPLES OF ROBOTICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Discuss in detail about any three types of robotic configurations with suitable sketches.	**08**
	b.	Explain the following terms: **i.** Repeatability **ii.** Accuracy **iii.** Spatial resolution with respect to working of robots.	**06**
2.	**a.**	Explain the significance of a homogenous transformation matrix as applied to a robotic manipulator.	**06**
	b.	With the help of a suitable diagram, define and explain D-H Parameters used for calculating forward kinematic metrics.	**08**
3.	**a.**	Define Principle of Virtual Work. Explain its significance with regards to robotic manipulators.	**06**
	b.	Derive an expression relating the Velocities of the Joints and end-effector of an RR manipulator.	**08**

OR

4.	**a.**	Discuss the singularity of robotic configurations? Derive the singularity condition for an RP manipulator	**09**
	b.	Explain in detail about 4R Chain Mechanism of Robotic Links with any two applications.	**05**
5.	**a.**	Define Trajectory Planning of robots? Explain the Cubic Polynomial equation for trajectory planning of robots.	**08**
	b.	A single-link robot with a revolute joint is motionless at $\Theta = 40^\circ$. It is desired to move the joint in a smooth manner to $\Theta = 80^\circ$ in 4.0 seconds. Find a suitable cubic polynomial to generate this motion and bring the manipulator to rest at the goal.	**06**

OR

6 a. Derive the Cubic Polynomial Function when initial and final values of joint angles (Θ) are known and angular velocities and accelerations at beginning and end of the cycle are assumed to have non zero values. **06**

b. Briefly explain the linear trajectory function with respect to trajectory planning for robots and its shortcoming with a suitable diagram. **08**

7. a. Derive an expression for Euler-Langrangian equation for calculation of torques in robotic link and explain its significance. **06**

b. Develop the matrix for moment of inertia tensor of a circular link in a robot. **08**

OR

8. a. Derive an equation for calculating the Kinetic energy of a robotic link. **07**

b. Deduce the equation for moment of inertia of a rectangular link with suitable assumptions. **07**